

TINKER AFB, OK, USA

WMO: 723540

Lat: 35.417N

Lon: 97.383W

Elev: 394

StdP: 96.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.5	-16.1	1.0	-7.5	-13.6	1.2	-4.3	13.5	5.9	12.4	5.8	5.2	0	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	37.8	23.0	36.1	23.2	34.4	23.4	25.6	32.3	25.1	31.8	24.5	31.2	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.7	28.6	23.0	18.7	28.0	22.6	18.2	27.6	81.1	32.3	78.3	31.7	76.0	31.3	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.9	9.5	DB	-13.8	39.4	2.1	2.2	-15.3	41.0	-16.5	42.3	-17.7	43.6	-19.2	45.2	(4)
(5)			WB	-14.4	26.9	1.9	1.1	-15.8	27.7	-16.9	28.3	-18.0	28.9	-19.4	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	4.0	6.3	11.1	15.8	20.6	25.3	27.9	27.5	23.3	16.9	10.5	5.1	(6)	
	DBStd	9.64	5.90	6.43	5.95	4.69	3.93	3.03	2.80	2.92	4.03	4.73	5.17	5.43	(7)	
	HDD10.0	631	200	134	57	9	1	0	0	0	0	6	56	167	(8)	
	HDD18.3	1879	446	338	231	101	24	1	0	0	8	82	238	410	(9)	
	CDD10.0	2908	13	31	92	182	329	458	553	543	401	221	70	16	(10)	
	CDD18.3	1114	0	1	7	24	94	209	295	285	158	38	3	0	(11)	
	CDH23.3	11615	1	11	65	196	748	2061	3499	3256	1465	291	22	1	(12)	
	CDH26.7	5400	0	2	12	38	218	886	1842	1700	630	71	2	0	(13)	
(14) Wind	WSAvg	4.8	5.2	5.2	5.6	5.7	4.9	4.6	4.1	3.9	4.0	4.7	5.1	4.9	(14)	
(15) Precipitation	PrecAvg	918	33	40	66	94	136	120	94	76	89	74	51	44	(15)	
	PrecMax	1353	84	105	166	180	411	286	222	195	225	258	156	145	(16)	
	PrecMin	608	0	0	9	19	39	26	15	1	21	9	0	0	(17)	
	PrecStd	194	22	27	42	44	87	65	59	47	46	53	45	32	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	24.8	28.2	30.1	32.9	37.5	40.1	40.4	38.0	31.1	25.9	21.3	(19)	
		MCWB	12.0	14.1	16.7	18.2	21.8	23.8	22.5	22.9	22.0	20.8	17.2	14.6	(20)	
	2%	DB	18.6	21.5	25.1	27.6	30.9	34.9	37.9	37.7	34.2	28.7	23.0	18.5	(21)	
		MCWB	11.7	13.0	15.9	18.2	22.0	23.6	23.1	23.0	22.3	19.7	15.6	13.3	(22)	
	5%	DB	16.0	18.7	22.6	25.7	29.1	32.9	36.1	36.1	32.4	26.6	20.9	16.0	(23)	
		MCWB	10.2	12.4	14.8	17.2	21.2	23.9	23.3	23.1	22.2	18.6	14.9	10.8	(24)	
	10%	DB	13.1	16.2	20.4	23.7	27.4	31.4	34.3	34.1	30.3	24.6	18.7	13.7	(25)	
		MCWB	8.1	11.1	13.9	16.4	20.6	23.5	23.7	23.1	21.5	17.9	13.9	9.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.0	17.4	19.4	21.9	24.7	26.4	26.6	26.3	24.6	22.9	19.6	17.6	(27)	
		MCDB	18.1	20.0	23.9	26.2	30.0	32.5	33.3	33.1	31.1	27.5	22.4	19.8	(28)	
	2%	WB	13.4	15.8	18.0	20.4	23.6	25.3	25.7	25.3	23.8	21.8	17.7	14.7	(29)	
		MCDB	16.2	18.7	22.4	24.3	28.3	31.4	32.5	32.5	30.2	26.0	20.8	16.6	(30)	
	5%	WB	10.8	13.8	16.8	19.4	22.7	24.7	25.1	24.7	23.1	20.7	16.4	12.2	(31)	
		MCDB	15.1	17.4	20.6	23.2	26.9	30.7	31.9	31.9	29.3	23.9	19.2	14.5	(32)	
	10%	WB	8.6	11.3	15.1	18.2	21.7	24.0	24.5	24.1	22.6	19.5	14.9	9.5	(33)	
		MCDB	12.7	16.2	19.3	22.0	25.9	29.8	31.3	31.3	28.5	22.8	17.9	13.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.4	11.7	11.5	10.4	10.0	11.1	11.3	10.9	11.0	11.0	10.3	(35)	
		MCDBR	15.4	15.1	14.6	13.8	11.8	11.7	13.5	14.0	13.0	13.1	13.7	13.8	(36)	
	5% WB	MCWBR	9.3	8.9	7.7	7.1	5.5	4.5	3.7	3.9	4.4	6.3	7.8	8.7	(37)	
		MCWBR	13.5	12.6	11.5	11.1	10.2	10.3	11.1	11.2	10.7	9.9	11.1	11.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.294	0.298	0.339	0.375	0.412	0.421	0.423	0.432	0.385	0.344	0.312	0.290	(40)	
		taud	2.507	2.512	2.421	2.333	2.299	2.315	2.341	2.257	2.370	2.478	2.506	2.525	(41)	
	Ebn at Noon		915	951	932	909	875	863	858	844	871	884	884	899	(42)	
		Edh at Noon	81	91	109	125	132	129	125	134	114	93	81	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.77	3.36	4.40	5.44	5.88	6.72	6.80	6.18	5.11	3.89	2.92	2.33	(44)	
	RadStd		0.28	0.51	0.28	0.40	0.63	0.65	0.36	0.38	0.46	0.50	0.38	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page